

Probability and Statistics

Chapter 4. Mathematical Expectation

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Chapter Overview

Mean of a Random Variable

Variance and Covariance of Random Variables

Means and Variances of Linear Combinations

Chebyshev's Theorem

Mean of a Random Variable

Mean of a Random Variable

Motivation

A random variable can take several possible values, each with a certain probability. To summarize the *center* or *long-run average* of this distribution, we use the **mean** or **expected value**.

- 기대값은 확률분포의 **가중평균**이다.
- 실제 관측값 중 하나가 아닐 수도 있다.
- 장기적으로 반복 시행할 때 나타나는 평균적인 수준을 의미한다.

Introductory Example: Two Coins

Suppose two fair coins are tossed once.

$$S = \{HH, HT, TH, TT\}$$

Let X be the number of heads. Then $X \in \{0, 1, 2\}$, and

$$P(X = 0) = \frac{1}{4}, \quad P(X = 1) = \frac{1}{2}, \quad P(X = 2) = \frac{1}{4}$$

Therefore,

$$E(X) = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = 1$$

Interpretation 두 동전을 한 번 던질 때 나오는 앞면 개수의 평균은 1이다.

Definition: Expected Value of X

Definition

Let X be a random variable with probability distribution $f(x)$. The mean, or expected value, of X is denoted by $\mu = E(X)$.

Discrete case

$$\mu = E(X) = \sum_x xf(x)$$

Continuous case

$$\mu = E(X) = \int_{-\infty}^{\infty} xf(x) dx$$

- discrete: 가능한 값마다 $x \times P(X = x)$ 를 더함
- continuous: $xf(x)$ 를 전체 구간에서 적분

Example: Number of Good Components

A lot contains 7 components: 4 good and 3 defective. A sample of 3 is selected without replacement.

Let X be the number of good components in the sample.

$$f(x) = \frac{\binom{4}{x} \binom{3}{3-x}}{\binom{7}{3}}, \quad x = 0, 1, 2, 3$$

Thus,

$$f(0) = \frac{1}{35}, \quad f(1) = \frac{12}{35}, \quad f(2) = \frac{18}{35}, \quad f(3) = \frac{4}{35}$$

Hence,

$$E(X) = 0 \cdot \frac{1}{35} + 1 \cdot \frac{12}{35} + 2 \cdot \frac{18}{35} + 3 \cdot \frac{4}{35} = \frac{12}{7} \approx 1.714$$

평균적으로 표본 3개에는 약 **1.7**개의 양품이 포함된다.

Expected Value of a Function $g(X)$

Theorem

For a random variable X with distribution $f(x)$, the expected value of $g(X)$ is

$$E[g(X)] = \sum_x g(x)f(x)$$

if X is discrete, and

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x) dx$$

if X is continuous.

Example: Continuous Random Variable

Suppose

$$f(x) = \begin{cases} \frac{20000}{x^3}, & x > 100, \\ 0, & \text{elsewhere} \end{cases}$$

Find the expected life of the device.

$$E(X) = \int_{100}^{\infty} x \cdot \frac{20000}{x^3} dx = \int_{100}^{\infty} \frac{20000}{x^2} dx$$

$$E(X) = 200$$

이 장치의 기대 수명은 **200**시간이다.

Expected Value of $g(X)$: Example

Let

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere} \end{cases}$$

and let $g(X) = 4X + 3$.

Then

$$E(4X + 3) = \int_{-1}^2 (4x + 3) \frac{x^2}{3} dx = \frac{1}{3} \int_{-1}^2 (4x^3 + 3x^2) dx$$

$$E(4X + 3) = 8$$

함수값의 기대값은 $g(E(X))$ 와 항상 같은 것은 아니다. 직접 $g(x)f(x)$ 를 이용해 계산해야 한다.

Expected Value of a Function of Two Variables

Definition

If X and Y have joint distribution $f(x, y)$, then

$$E[g(X, Y)] = \sum_x \sum_y g(x, y) f(x, y)$$

for discrete random variables, and

$$E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) dx dy$$

for continuous random variables.

Example: $E(XY)$

For discrete random variables X and Y ,

$$E(XY) = \sum_x \sum_y xy f(x, y)$$

- joint distribution이 주어지면 각 셀에 대해 $xyf(x, y)$ 를 더한다.
- $x = 0$ 또는 $y = 0$ 인 항은 $xy = 0$ 이므로 기여하지 않는다.

이중합의 핵심은 각 조합 (x, y) 에 대해

(함수값) $\times$ (확률)

을 더하는 것이다.

Variance and Covariance of Random Variables

Variance and Dispersion

Key Idea

Two distributions can have the same mean but very different spreads. The mean describes the center, while the **variance** describes the dispersion.

- 평균이 같아도 흩어진 정도는 다를 수 있다.
- 분산은 평균으로부터 얼마나 퍼져 있는지를 수치화한다.
- 표준편차는 분산의 양의 제곱근이다.

Definition of Variance

Definition

If X has mean μ , then the variance of X is

$$\sigma^2 = E[(X - \mu)^2]$$

Discrete case

$$\sigma^2 = \sum_x (x - \mu)^2 f(x)$$

Continuous case

$$\sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$

The **standard deviation** is

$$\sigma = \sqrt{\sigma^2}$$

Example: Equal Means, Different Variances

For company A:

$$x : 1, 2, 3 \quad f(x) : 0.3, 0.4, 0.3$$

For company B:

$$x : 0, 1, 2, 3, 4 \quad f(x) : 0.2, 0.1, 0.3, 0.3, 0.1$$

Both have

$$E(X) = 2$$

But

$$\sigma_A^2 = (1-2)^2(0.3) + (2-2)^2(0.4) + (3-2)^2(0.3) = 0.6$$

$$\sigma_B^2 = (0-2)^2(0.2) + (1-2)^2(0.1) + (2-2)^2(0.3) + (3-2)^2(0.3) + (4-2)^2(0.1) = 1.6$$

Shortcut Formula for Variance

Theorem

$$\text{Var}(X) = \sigma^2 = E(X^2) - \mu^2$$

where $\mu = E(X)$.

Why useful?

- 직접 $E[(X - \mu)^2]$ 를 계산하는 대신
- 먼저 $E(X)$, $E(X^2)$ 를 구한 뒤
- $\text{Var}(X) = E(X^2) - [E(X)]^2$ 를 사용하면 편리하다.

Example: Variance by Shortcut Formula

Suppose

$$x : 0, 1, 2, 3 \quad f(x) : 0.51, 0.38, 0.10, 0.01$$

Then

$$E(X) = 0(0.51) + 1(0.38) + 2(0.10) + 3(0.01) = 0.61$$

and

$$E(X^2) = 0^2(0.51) + 1^2(0.38) + 2^2(0.10) + 3^2(0.01) = 0.87$$

Therefore,

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 0.87 - (0.61)^2$$

$$\text{Var}(X) = 0.4979$$

이 예제는 $\text{Var}(X) = E(X^2) - \mu^2$ 공식이 실제 계산에서 매우 유용함을 보여준다.

Covariance: Basic Idea

Covariance

Covariance measures how two random variables vary together.

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Equivalent form:

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

- $\text{Cov}(X, Y) > 0$: 함께 커지는 경향
- $\text{Cov}(X, Y) < 0$: 하나가 커질 때 다른 하나는 작아지는 경향
- $\text{Cov}(X, Y) = 0$: 선형적인 함께 변동하는 경향이 없음

Covariance: Equivalent Form

Theorem

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E(XY) - E(X)E(Y)$$

Proof

$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - \mu_X)(Y - \mu_Y)] \\&= E[XY - \mu_X Y - \mu_Y X + \mu_X \mu_Y] \\&= E(XY) - \mu_X E(Y) - \mu_Y E(X) + \mu_X \mu_Y \\&= E(XY) - \mu_X \mu_Y - \mu_Y \mu_X + \mu_X \mu_Y \\&= E(XY) - \mu_X \mu_Y \\&= E(XY) - E(X)E(Y)\end{aligned}$$

Properties of Covariance

Basic Properties

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

$$\text{Cov}(X, X) = \text{Var}(X)$$

$$\text{Cov}(X, c) = 0 \quad (c : \text{constant})$$

$$\text{Cov}(aX + b, Y) = a \text{Cov}(X, Y)$$

$$\text{Cov}(X, cY + d) = c \text{Cov}(X, Y)$$

$$\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$$

- covariance는 두 변수의 **함께 변하는 정도**를 측정한다.
- 상수항 b, d 는 covariance에 영향을 주지 않는다.
- 배율 a, c 는 covariance의 크기를 비례적으로 바꾼다.

Covariance and Independence

Important Fact

If X and Y are independent, then

$$E(XY) = E(X)E(Y)$$

so

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 0$$

Warning

$$\text{Cov}(X, Y) = 0$$

does **not** always imply independence.

Why $\text{Cov}(X,Y)=0$ does NOT imply independence

Key Idea

$$\text{Cov}(X, Y) = 0$$

means **no linear relationship**, not independence.

- Covariance는 두 변수의 **선형적인 함께 변동**만 측정한다.
- 따라서 비선형 관계는 covariance로 감지되지 않는다.

Conclusion

$$\text{Cov}(X, Y) = 0 \not\Rightarrow X \text{ and } Y \text{ are independent}$$

Example: Uncorrelated but Not Independent

Let $X \in \{-1, 0, 1\}$ with

$$P(X = -1) = P(X = 0) = P(X = 1) = \frac{1}{3}$$

and define

$$Y = X^2$$

Then

$$E(X) = (-1) \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} = 0$$

$$E(Y) = E(X^2) = 1 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} = \frac{2}{3}$$

$$E(XY) = E(X^3) = (-1)^3 \frac{1}{3} + 0^3 \frac{1}{3} + 1^3 \frac{1}{3} = 0$$

Example: Interpretation

Therefore,

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 0 - 0 \cdot \frac{2}{3} = 0$$

Interpretation

Although

$$\text{Cov}(X, Y) = 0,$$

the variable $Y = X^2$ is completely determined by X . So X and Y are **not independent**.

- covariance가 0이라는 것은 **선형적 관계가 없다**는 뜻에 가깝다.
- 완전한 독립과는 다른 개념이다.

Example: Covariance in a Sensor System

Suppose two sensor errors (X, Y) have the joint pmf

	$Y = -1$	$Y = 1$
$X = -1$	0.3	0.2
$X = 1$	0.1	0.4

Then

$$E(X) = (-1)(0.3 + 0.2) + (1)(0.1 + 0.4) = 0$$

$$E(Y) = (-1)(0.3 + 0.1) + (1)(0.2 + 0.4) = 0.2$$

$$E(XY) = 1(0.3) + (-1)(0.2) + (-1)(0.1) + 1(0.4) = 0.4$$

Example: Result and Meaning

Hence,

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 0.4 - 0 \cdot 0.2 = 0.4$$

Interpretation

Since

$$\text{Cov}(X, Y) > 0,$$

the two sensor errors tend to move in the same direction.

- 한 센서 오차가 커질 때 다른 센서 오차도 함께 커지는 경향이 있다.
- 이런 정보는 센서 융합(sensor fusion)이나 공통 잡음 분석에 유용하다.

Covariance in Variance Formulas

Useful Identities

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X, Y)$$

$$\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X, Y)$$

- 기대값과 달리 분산에는 covariance 항이 추가된다.
- 두 변수가 함께 움직이면 합이나 차의 분산이 달라진다.

Example: Combined Measurement Error

Suppose

$$\text{Var}(X) = 4, \quad \text{Var}(Y) = 9, \quad \text{Cov}(X, Y) = 3$$

Then

$$\text{Var}(X + Y) = 4 + 9 + 2(3) = 19$$

and

$$\text{Var}(X - Y) = 4 + 9 - 2(3) = 7$$

Interpretation

When covariance is positive, subtraction can reduce variability. This idea appears in differential measurement and noise cancellation.

Means and Variances of Linear Combinations

Linear Combinations

In many applications, we study

$$Y = aX + b \quad \text{or} \quad W = a_1X_1 + a_2X_2 + \cdots + a_nX_n$$

Question

If we know the means and variances of the original random variables, how do we find the mean and variance of the new variable?

Mean of a Linear Function

Theorem

For constants a and b ,

$$E(aX + b) = aE(X) + b$$

- 기대값은 선형성(linearity)을 가진다.
- 상수배는 밖으로 나오고, 상수항은 그대로 더해진다.

Example:

$$E(4X + 3) = 4E(X) + 3$$

Variance of a Linear Function

Theorem

For constants a and b ,

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

- 상수 b 를 더해도 분산은 변하지 않는다.
- 상수 a 배를 하면 분산은 a^2 배가 된다.

In particular,

$$\text{Var}(X + b) = \text{Var}(X)$$

$$\text{Var}(aX) = a^2 \text{Var}(X)$$

Variance of a Sum

For two random variables X and Y ,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

Similarly,

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X, Y)$$

Special Case: Independence

If X and Y are independent, then

$$\text{Cov}(X, Y) = 0$$

so

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

Mean of a Linear Combination

For

$$W = a_1X_1 + a_2X_2 + \cdots + a_nX_n$$

we have

$$E(W) = a_1E(X_1) + a_2E(X_2) + \cdots + a_nE(X_n)$$

기대값은 항상 선형적이므로, 독립 여부와 관계없이 그대로 분리된다.

Variance of a Linear Combination

For

$$W = a_1X_1 + a_2X_2 + \cdots + a_nX_n$$

the variance is

$$\text{Var}(W) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{i < j} a_i a_j \text{Cov}(X_i, X_j)$$

If $X_1, \dots, X_n$ are independent, then all covariance terms vanish:

$$\text{Var}(W) = \sum_{i=1}^n a_i^2 \text{Var}(X_i)$$

분산은 기대값과 달리 **공분산 항**이 추가된다는 점이 핵심이다.

Example 1: Linear Function

Suppose

$$E(X) = 5, \quad \text{Var}(X) = 4$$

and

$$Y = 3X - 2$$

Question

Find $E(Y)$ and $\text{Var}(Y)$.

Solution

$$E(Y) = 3E(X) - 2 = 3(5) - 2 = 13$$

$$\text{Var}(Y) = 3^2 \text{Var}(X) = 9(4) = 36$$

Example 2: Independent Sum

Suppose X, Y are independent:

$$E(X) = 2, \quad \text{Var}(X) = 1$$

$$E(Y) = 4, \quad \text{Var}(Y) = 9$$

Let $W = X + Y$.

Question

Find $E(W)$ and $\text{Var}(W)$.

Solution

$$E(W) = 2 + 4 = 6$$

$$\text{Var}(W) = 1 + 9 = 10$$

Example 3: Correlated Variables

Suppose

$$E(X) = 10, \quad \text{Var}(X) = 5$$

$$E(Y) = 6, \quad \text{Var}(Y) = 3$$

$$\text{Cov}(X, Y) = 2$$

Let $Z = X - Y$.

Question

Find $E(Z)$ and $\text{Var}(Z)$.

Example 3: Solution

Solution

$$E(Z) = 10 - 6 = 4$$

$$\text{Var}(Z) = 5 + 3 - 2(2) = 4$$

Example 4: Weighted Sum

Let independent X_1, X_2, X_3 :

$$E(X_1) = 1, E(X_2) = 2, E(X_3) = 3$$

$$\text{Var}(X_1) = 2, \text{Var}(X_2) = 1, \text{Var}(X_3) = 4$$

$$W = 2X_1 - X_2 + 3X_3$$

Question

Find $E(W)$ and $\text{Var}(W)$.

Example 4: Solution

Solution

$$E(W) = 2(1) - 2 + 3(3) = 9$$

$$\text{Var}(W) = 4(2) + 1(1) + 9(4) = 45$$

Example 5: Sensor Combination

Suppose

$$E(X) = 50, \quad \text{Var}(X) = 16$$

$$E(Y) = 30, \quad \text{Var}(Y) = 9$$

$$\text{Cov}(X, Y) = 6$$

$$S = 0.5X + 2Y$$

Question

Find $E(S)$ and $\text{Var}(S)$.

Example 5: Solution

Solution

$$E(S) = 0.5(50) + 2(30) = 85$$

$$\text{Var}(S) = 0.25(16) + 4(9) + 2(6) = 52$$

Chebyshev's Theorem

Chebyshev's Theorem

Theorem

For any random variable X with mean μ and standard deviation σ , the probability that X lies within k standard deviations of the mean is at least

$$1 - \frac{1}{k^2}, \quad k > 1$$

Equivalently,

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$$

Interpretation of Chebyshev's Theorem

- 분포의 모양이 무엇이든 항상 성립한다.
- 정규분포가 아니어도 적용 가능하다.
- 따라서 매우 일반적인 하한(lower bound)을 제공한다.

Examples:

$$k = 2 \Rightarrow P(|X - \mu| < 2\sigma) \geq 1 - \frac{1}{4} = 0.75$$

$$k = 3 \Rightarrow P(|X - \mu| < 3\sigma) \geq 1 - \frac{1}{9} = \frac{8}{9} \approx 0.889$$

적어도 75%의 값은 평균에서 2표준편차 이내에 있고, 적어도 88.9%의 값은 평균에서 3표준편차 이내에 있다.

Why Chebyshev's Theorem Matters

- 분포의 정확한 형태를 몰라도 확률을 하한으로 제어할 수 있다.
- 데이터가 얼마나 평균 주변에 모여 있는지 설명할 수 있다.
- 확률론과 통계학에서 매우 기본적인 부등식이다.

Core Message

Mean tells us the *center*, variance tells us the *spread*, and Chebyshev's theorem links spread to probability.

Example: Chebyshev's Inequality

Suppose a random variable X has

$$\mu = 100, \quad \sigma = 10$$

Find a lower bound for

$$P(80 < X < 120)$$

Solution

$$80 < X < 120 \iff |X - 100| < 20$$

$$|X - \mu| < 2\sigma$$

By Chebyshev's inequality,

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$$

Example: Chebyshev's Inequality

with $k = 2$,

$$P(|X - 100| < 20) \geq 1 - \frac{1}{4} = 0.75$$

Interpretation

At least 75% of the values lie between 80 and 120, regardless of the distribution shape.